Fatima Salahuddin

LinkedIn: https://www.linkedin.com/in/fatima-salahuddin-/
Phone: +92 3146682902 | Email: fatimasalahuddinmirza@gmail.com

Computer Science undergraduate passionate about practical problem-solving, with growing experience in software development, systems programming, and cross-functional collaboration.

EDUCATION

National University of Computer and Emerging Sciences (NUCES, FAST) **Karachi, Pakistan**
Bachelor of Science in Computer Science
Rector's List, Gold Medalist, 100% merit scholarship 2024 - 2027

BAMM PECHS Govt. College **Karachi, Pakistan**
High School in Pre-Engineering
3rd position BIEK Intermediate Board 2021 – 2023

WORK & LEADERSHIP EXPERIENCE

Digital Literacy Instructor **Tabuk, Saudi Arabia (Remote)**
Consultant Jan 2024 – Oct 2025
Delivered a comprehensive ICT and digital literacy curriculum for students aged 8–17, training over 30 students in computer fundamentals, internet safety, digital communication, and design tools, while creating more than 10 interactive lesson plans, activities, and project-based modules.

Private Tutor – CS, Python & Math **Karachi, Pakistan**
Self-Employed Jun 2023 – Present
Tutored over 50 students in programming logic, ICT, and mathematics, employing personalized strategies to enhance academic performance, while maintaining regular communication with parents and tracking each student's progress.

PROJECTS

- **E-Commerce Website Database System (React + Node.js + Flask + Oracle DB)**

Developed a full-stack e-commerce platform with authentication, product management, and CRUD operations, integrating React frontend, Node.js & Flask backend, and Oracle DB.

- **AI Project – GAN Model (TensorFlow + Streamlit)**

Developed a Generative Adversarial Network (GAN) using TensorFlow with a **double-headed discriminator** to improve output evaluation, and built a Streamlit dashboard for interactive visualization. Optimized the training loop, evaluated performance, and implemented the complete end-to-end workflow for synthetic data generation.

- **Networks & Algorithms Project**

Designed a network load balancer for efficient traffic distribution and implemented optimized algorithms with performance analysis.

- **Matrix Calculator**

Developed a matrix calculator in x86 Assembly Language using Irvine32 with menu-driven operations like addition, multiplication, and determinant calculation.

- **Advanced Producer-Consumer Kernel Module**

Developed a Linux kernel module implementing a priority-based producer-consumer system using kernel threads, semaphores, and spinlocks. Simulated interrupt handling and deferred work using IRQ and workqueues; enabled real-time configuration via sysfs and debugfs.

EXTRACURRICULAR ACTIVITIES

- Head, Graphic Design – Robotics and Automation Society; Head, Design – Developer Student Club; Co-Head of Marketing – FAST Entrepreneurship Society; Co-Head, Design – ACM -W
- Taught English at Govt. Polytechnic Institute, Women; donation drives with HOPES to provide free medical services raising Rs.67K+; visited orphanages and interacted with children like Dar-ul-Sukoon, Farzana Orphanage, etc.

SKILLS, ACTIVITIES & INTERESTS

Technical Skills: C/C++, Python, SQL, Linux, Bash, Assembly Language, HTML, CSS, React, Node.js, Rest APIs, GIT, DSA, OOP, Photoshop, Illustrator, Canva, MongoDB
Certifications & Training: Google AI Essentials, AI Seekho (Google Cloud – AI & ML Training), C Programming Fundamentals & Structured Programming (University of California, Santa Cruz), Python for Beginners, HTML & CSS
Interests: Graphic Design, Content Writing, Community Outreach, F1, AI, Exploring New Ideas